Project Design Phase

Solution Architecture

Date	08 Feb 2026
Team ID	LTVIP2026TMIDS84828
Project Name	Financial AI Analyzer – Transforming Complex Data into Actionable Insights
Maximum Marks	4 Marks

Solution Architecture

1. Overview

The Financial AI Analyzer Solution Architecture is designed as a cloud-based, AI-driven financial data analysis system that integrates data processing, artificial intelligence, and interactive dashboards to generate meaningful financial insights.

The architecture bridges the gap between complex financial datasets and intelligent decision-making by combining a web interface, financial data processing modules, AI analysis engines, and visualization components.

The system follows a modular layered architecture consisting of:

- Presentation Layer (User Interface)
- Application Layer (Business Logic)
- Data Processing Layer (Financial Data Analysis)
- AI Processing Layer (AI Insight Generation)
- Session & Temporary Storage Layer
- Cloud Infrastructure Layer

This architecture ensures scalability, reliability, maintainability, and efficient financial data processing.

2. Architecture Components

2.1 User Interface Layer (Presentation Layer)

Technology: Streamlit (Python Web Framework)

Responsibilities

- Accept financial dataset uploads (CSV / Excel)
- Accept financial queries from users
- Display financial dashboards and charts
- Display AI-generated insights and summaries
- Manage switching between Data Analysis Mode and AI Assistant Mode
- Show analysis history during the session

2.2 Application Logic Layer

Technology: Python

Modules

- Data Validation & Preprocessing Module
- Financial Data Processing Module
- Prompt Engineering Module
- Query Processing Module
- Response Formatting Module
- Error Handling Module
- Session State Management

Responsibilities

- Validate uploaded datasets
- Process financial data for analysis
- Construct structured prompts for AI
- Format AI-generated financial insights
- Maintain user session and history

2.3 Data Processing Layer

Technology: Pandas / NumPy

Responsibilities

- Clean and preprocess financial datasets
- Transform raw data into structured formats
- Calculate financial metrics and indicators
- Identify financial trends and patterns

This layer prepares financial data before sending it to the AI processing module.

2.4 AI Processing Layer

Technology: Generative AI / Gemini API

Responsibilities

- Financial data interpretation
- Trend analysis and forecasting
- Risk analysis
- Investment insights generation
- AI-based financial recommendations

This layer acts as the core intelligence engine of the system.

2.5 Session & Temporary Storage Layer

Data Stores

D1 – Session State

- Stores financial analysis history
- Stores chat interactions with AI assistant
- Maintains user interface mode

D2 – Temporary Dataset Storage

- Holds uploaded datasets temporarily during analysis

Currently, the system does not require a permanent database.

2.6 Cloud Infrastructure Layer

Deployment Options

- Streamlit Cloud
- Local Server
- Cloud Virtual Machine (AWS / GCP / Azure)

API Communication

Secure API key management via:

• Streamlit Secrets (Cloud)
• .env file (Local Environment)

3. End-to-End Data Flow

Financial Data Analysis Flow

1. User uploads financial dataset
2. Dataset is validated and preprocessed
3. Structured financial data sent to AI processing module
4. AI analyzes data and generates insights
5. Application formats results into charts and summaries
6. Financial dashboard displayed to the user

Conversational AI Assistant Flow

1. User enters financial query
2. Query converted into structured AI prompt
3. Prompt sent to AI processing module
4. AI generates contextual financial response
5. Response stored in session state
6. Chat displayed with conversation history

4. Architectural Characteristics

4.1 Modular Design

Clear separation between UI, processing logic, AI engine, and storage layers.

4.2 Scalable

AI processing handled through cloud-based AI APIs, allowing the system to scale with increasing data volume and users.

4.3 Secure

Sensitive information such as API keys managed using environment variables and secure configuration files.

4.4 Maintainable

Financial data processing and AI modules are independent from the UI layer, making updates easier.

4.5 Extensible

Future enhancements may include:

- Real-time financial data integration
- Predictive financial modeling
- Database integration
- Multi-language support
- Advanced financial risk analysis tools

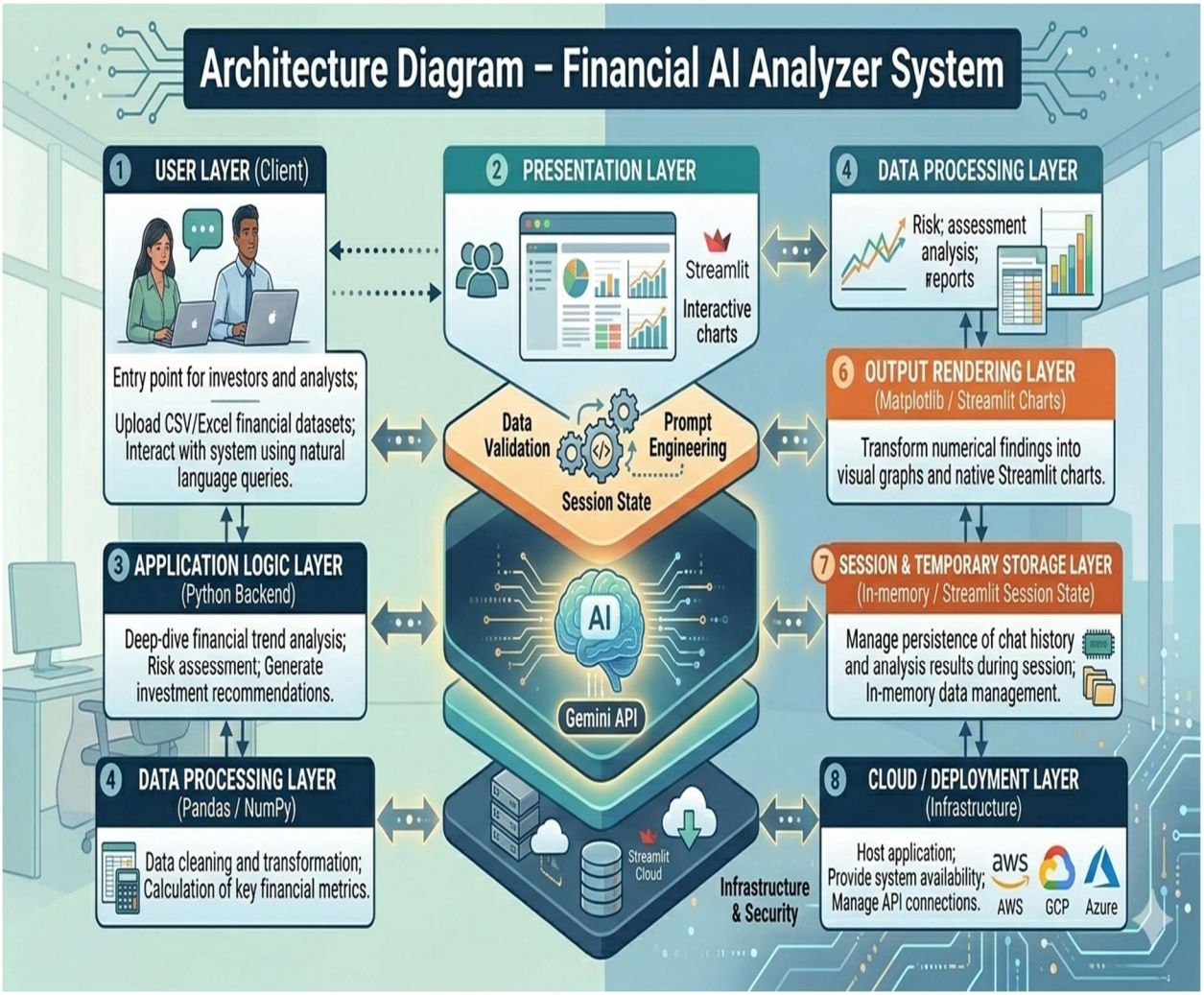
5. Architecture Style

The system follows:

- Layered Architecture
- API-Centric Architecture
- Cloud-Based AI Integration
- Stateless Application with Session Memory

6. Solution Architecture Summary

The Financial AI Analyzer architecture transforms complex financial data into intelligent AI-driven insights through a scalable and modular system design. By combining AI analysis, financial data processing, and interactive dashboards, the platform enables users to quickly interpret financial datasets and make informed decisions.

The solution ensures high usability, scalability, and maintainability, while delivering real-time financial insights through an intuitive web interface.

Solution Architecture Diagram: